NADINE EL-MUFTI

Canadian Citizen, residing in Montreal, QC
(514) 802-3932 | | | |

EXECUTIVE SUMMARY

Machine learning researcher and data scientist specializing in healthcare AI, with experience developing end-to-end NLP and speech-processing pipelines for neurological and psychiatric applications. Skilled in Python-based machine learning across the full lifecycle, including dataset curation, clinical data engineering, statistical and confounder analyses, model development, evaluation, optimization, and interpretability. Also proficient in Excel for data analysis and reporting. Active researcher in responsible AI and AI governance, examining fairness, ethical pluralism, and how stakeholder values shape the governance and clinical deployment of healthcare AI through participatory and anticipatory approaches.

WORK EXPERIENCE

The Douglas Research Centre, McGill University — Montreal, QC
Graduate Research Assistant — Jun 2025 – Present

- Advancing speech-based relapse prediction in psychotic disorders as a data scientist, through multisite data collection, longitudinal speech data engineering pipelines, and rigorous statistical analyses investigating associations between speech-derived features, symptom severity, cognition, and demographic factors across clinical cohorts.

Mila - Quebec AI Institute — Montreal, QC
Collaborating Researcher / Master's Research — Oct 2024 – Present

- Leading research on speech as a biomarker for the detection and longitudinal monitoring of Parkinson's disease, covering the full machine learning lifecycle from dataset curation and clinical data integration to model training, evaluation, and optimization, while investigating confounding factors across in-the-wild and clinical recordings, and applying participatory foresight to guide responsible clinical deployment.

Applied Perception Lab, Concordia University — Montreal, QC
Graduate Research Assistant / Master's Research — Dec 2024 – Present

- Leading research on responsible and ethical healthcare AI, spanning anticipatory fairness, participatory foresight, and AI governance, by developing a stakeholder-informed, SDG-aligned ten-principle framework to evaluate future deployment scenarios with cross-disciplinary raters, surface value conflicts early, and identify distinct moral orientations that strengthen governance committees through moral diversity beyond demographics.

Concordia University — Montreal, QC
Teaching Assistant — Jan 2025 – April 2025

- Achieving consistent and fair grading for Concordia's Machine Learning course 'Comp 432' by marking labs, delivering prompt feedback, and addressing inquiries from 110 students.

SKILLS

Python (Advanced) · SQL (Advanced) · R (Intermediate) · pandas (Advanced) · NumPy (Advanced) · Jupyter (Advanced) · REDCap (Advanced) · Statistical Analysis (Intermediate) · scikit-learn (Advanced) · PyTorch (Advanced) · Hugging Face (Intermediate) · React (Intermediate) · JavaScript (Intermediate) · Figma (Intermediate) · Adobe Creative Suite (Advanced) · Microsoft Excel (Advanced) · LaTeX (Advanced) · Linux & Windows (Intermediate) · Git/GitHub (Advanced) · HPC/SLURM (Intermediate) · Research · Grant Writing

PUBLISHED & ON-GOING RESEARCH

- **El-Mufti, N.**, Mackinley, M., Dzialoszynski, P., Palaniyappan, L., Voppel, A.* Automatic Speech Recognition Performance in Psychiatric Speech: Linguistic, Clinical, and Architectural Factors *(Submitted to NPP—Digital Psychiatry and Neuroscience (DPN), Nature Portfolio).*
- **El-Mufti, N.**, Drouin, S., Jannin, P., Kersten-Oertel, M.* Futures Before Failures: Design Fictions for Anticipatory Fairness in Surgical AI. *(Accepted at the Joint FAIMI-BRIDGE-EPIMI Workshop, MICCAI 2026).* https://papers.miccai.org/miccai-2026-sat/paper/FAIMI-BRIDGE-EPIMI_029.pdf.
- **El-Mufti, N.**, Kersten-Oertel, M., Ravanelli, M.* Enhancing Speech-Based Parkinson's Detection with Celebs4PD: A Corpus of Natural, In-the-Wild, Longitudinal Celebrity Speech. *(In Preparation).*
- El-Mufti, N., Ravanelli, M., Kersten-Oertel, M.* From Speech Biomarkers to Clinical Practice: Participatory Foresight for Responsible Clinical Deployment in Parkinson's Disease. *(In Preparation).*
- Reyes, J., **El-Mufti, N.**, Gorman, S., Xie, D., Kersten-Oertel, M.* User-Centered Design for Surgical Innovations: A Ventriculostomy Case Study. EPIMI Workshop, MICCAI 2022, LNCS vol. 13755, 2022. https://doi.org/10.1007/978-3-031-23223-7_5.

EDUCATION

Master of Computer Science | Computer Science — Montreal, Quebec
Concordia University / Mila - Quebec AI Institute — Jan 2025 – Present

Bachelor of Engineering | Software Engineering — Montreal, Quebec
GPA (3.67) | With Distinction — Jan 2019 – May 2024
Concordia University

SCHOLARSHIPS, HONORS & AWARDS

Power Corporation of Canada Graduate Fellowship (Concordia University, Jun 2026, CAD 5,000) · Arbour Foundation Master's Scholarship (Arbour Foundation, Sep 2025, CAD 43,000) · FRQNT B1X Award Competition – 92/100, Ranked 9th (Funding Cutoff: 8) (FRQNT, Apr 2025) · Special Entrance Award (Concordia University, Jan 2025, CAD 6,000) · NSERC CGS-M Award (NSERC, Jan 2025, CAD 27,000) · Bourses d'Excellence for Engineering Scholarship (Quebec Ministry of Education, Jun 2022, CAD 3,000) · Supplements of the NSERC Undergraduate Student Research Awards - USRA (BPCA) (FRQNT, Dec 2022, CAD 1,500) · NSERC Undergraduate Student Research Award (USRA) (NSERC, May 2022, CAD 6,000)

LEADERSHIP, SERVICE & ENGAGEMENT

The International Association for Safe & Ethical AI (Member, Sep 2026 – Present) · Arabs in Neuroscience (Member, Apr 2026 - Present) · The Levantines (Member, Feb 2026 - Present) · Association Étudiante de Mila (Member, Oct 2025 – Present) · Arbour Foundation Scholars (Member, Sep 2025 – Present) · Golden Key International Honour Society (Member, Sep 2025 – Present) · Brainhack MTL | Brainhack Global (Organizer, Jul 2025 – Present) · Graduate Students' Association of Concordia GSA (Subcommittee Member, Jan 2025 – Apr 2025) · School of Graduate Studies, Concordia University (Ambassador, Sep 2024 – May 2026) · GEMinAI | Concordia AI2 (Mentee, Oct 2024 – May 2025)

LICENSES & CERTIFICATIONS

Transatlantic Summer School (UdeM x LMU, Jun 2026) · Master AI Tools for Research (WiseUp Communications, Jun 2026) · Foundations in Responsible AI & AI Ethics *(Mila, May 2026)* · Bell x Mila x Buzz HPC x Kids Help Phone Hackathon *(Mila, Apr 2026)* · Ethics and Governance of AI for Health (WHO Academy, Jan 2026) · QcSE Program (V1 Studio, Jun 2025) · Strategic Public Communications (GradProSkills, Mar 2025) · TRAIL AI Learning Certificate (Mila, Mar 2025) · TCPS 2: CORE 2022 (Government of Canada, Jan 2025) · Tech Stewardship Certification (Tech Stewardship Network, Jul 2024)
The Governance of AI – Thematic Semester (IVADO, Aug–Oct 2026) | Workshops 1 · 2